

Experiment No. 5

Title:

Demonstration of Array Methods and Object Handling; Development of a Shopping Cart Calculator with Discount Logic using JavaScript.

Software / Tools Required:

1. Visual Studio Code (VS Code)
2. Google Chrome
3. HTML5
4. CSS3
5. JavaScript (ES6)

Theory:

Arrays and objects are important data structures in JavaScript used to store and manage collections of data.

(a) Array Methods

JavaScript provides various methods to perform operations on arrays, such as:

1. `push()` – adds an element to the end of an array.
2. `forEach()` – executes a function for each element.
3. `map()` – creates a new array by applying a function to each element.
4. `filter()` – creates a new array containing elements that satisfy a condition.
5. `reduce()` – reduces an array to a single value.

(b) Objects

Objects store data in the form of **key-value pairs**. In this practical, product details such as name, price, and quantity are stored as objects.

(c) Shopping Cart Calculator

A shopping cart is represented using an array of product objects. Array methods are used to display products, calculate the total bill, and generate item summaries.

(d) Discount Logic

Discount logic calculates the discount according to the total cart value. The discount amount is subtracted from the total price to obtain the final payable amount.

Shopping Cart – Total Calculator

Accepts User Input Such As:

1. Product Name
2. Price
3. Quantity

It Calculates:

1. Individual Product Total
2. Cart Total
3. Discount
4. Final Amount

It Also Performs:

1. Adds products using `push()`.
2. Displays products using `forEach()`.
3. Calculates the total using `reduce()`.
4. Generates item summaries using `map()`.
5. Filters expensive products using `filter()`.

Result / Conclusion:

The practical was completed successfully. The concepts of **array methods and object handling** were implemented. A shopping cart calculator was developed using JavaScript that calculates the total cart value, applies the appropriate discount, and displays the final payable amount accurately.